CCCA Cell-State Analysis — Wang et al. 2019

Glioblastoma (Brain) · 10x scRNA-seq · 9,632 malignant cells · 7 samples

GO / NO-GO GATE LEDGER

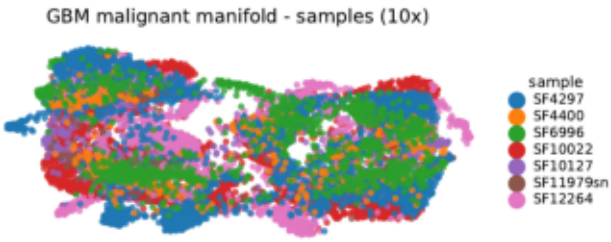
GO **Gate 1**
Samples intermix on one continuous connected manifold. Per-neighborhood iLISI rises 1.03 (pre-Harmony) to 2.60 (theta=10) and plateaus; n_pcs 20/40 near-identical (not under-powered). One component after island removal.

GO **Gate 2**
Embryonic module pattern present (non-branching). Gast/pSom co-vary spatially (r=+0.16) along one axis and both oppose Crest (Gast r=-0.49, pSom r=-0.87); Crest is the inverse terminal gradient (high-decile pseudotime 0.84).

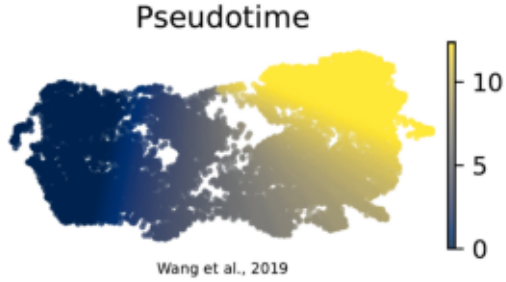
NA **Gate 3**
Not applicable — no bifurcation. Gast and pSom co-vary rather than cross-repress, so there is no Gast/pSom hub diverging into two competing arms to test. Geometry is linear (curve: 2 tips, 0 forks, fold-free ratio 2.82).

STEP FIGURES

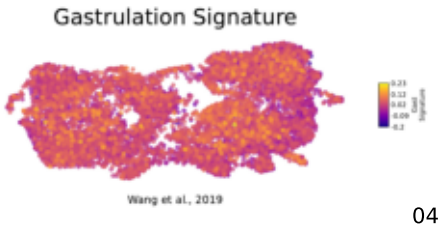
01 · Samples (integration)



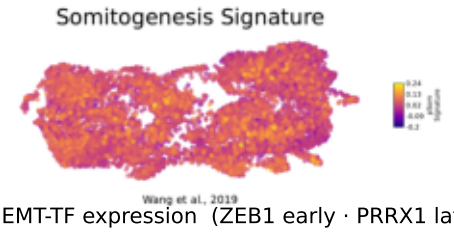
02 · Manifold (pseudotime)



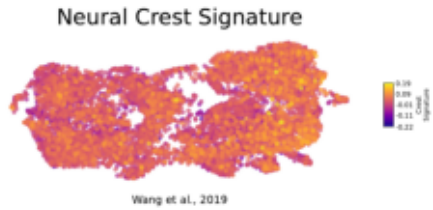
03 · Gastrulation



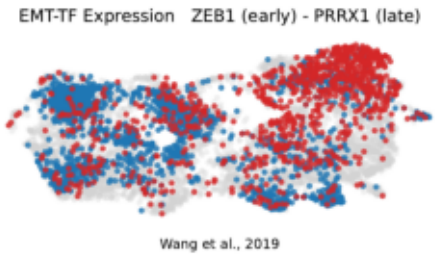
03 · Somitogenesis



03 · Neural Crest



04 · EMT-TF expression (ZEB1 early · PRRX1 late)



KEY PARAMETER CHOICES & RATIONALE

- **Min-cell threshold:** 237 (7/10 samples kept, 9,916 → 9,650 cells) — largest ratio gap in sorted per-sample counts (396→143, ratio 2.77), cut at the geometric mean.
- **Integration:** Harmony on sample, theta=10, n_pcs=20 — mixing plateaus (iLISI 2.4→2.6, theta 5→10); n_pcs in the robust 20–40 range; no sc.pp.scale before PCA.
- **Island removal:** UMAP radius graph (6× median NN), keep largest component — 18 cells in 2 detached specks removed → 9,632 cells on one component.
- **Subtypes:** AC-like / OPC-like / NPC-like — data marker scores per Leiden cluster, cross-referenced to the publication meta-programs (obs.mp_assignment).
- **Geometry / root:** Linear (non-branching) scFates curve; root at the Gast/pSom-high, Crest-low developmental origin tip; continuity +0.98 on an independent diffusion embedding.

VERDICT

The embryonic program reproduces as a single collapsed (linear) axis: a Gast/pSom-high developmental origin grading to a Crest-high terminus, EMT-TFs ZEB1 (early) → PRRX1 (late). No Gast/pSom bifurcation — Gate 3 NA. Gate 1 GO, Gate 2 GO (non-branching).

Gast μ = −0.002 pSom μ = +0.008 Crest μ = −0.007